

Engage a modern methodology to aid the design of a system from initial requirements

1. *Outline the project management strategy the project will use*
2. *Establish the requirement for the project*
 - a. *Human*
 - b. *Hardware*
 - c. *Software*
3. *Use of correct documentation and content*

1. Outline the project management strategy the project will use:

1. **Establish project objectives and scope:**

- The first step in any project is to clarify the overall purpose, and goals of the project, as well as outline the specific activities that will be completed. This should include a clear definition of the scope of the project, budget, timeline, and any other constraints.

2. **Define roles and responsibilities:**

- It is important to define who is responsible for different aspects of the project and make sure everyone understands their role and responsibility. This includes both internal and external stakeholders, such as contractors and vendors.

3. **Develop a project plan:**

- A project plan is essential for any successful project, and should include milestones, tasks, deliverables, and target dates. It should also define how the project will be monitored and adjusted if necessary.

4. **Identify resources:**

- Resources are an essential component of any project. Identifying the right people and other resources, such as hardware and software, is critical to success.

5. **Manage the project timeline:**

- Once the project plan has been developed, it is important to monitor progress and ensure that deadlines are met. This involves managing tasks, resources, and communication with all stakeholders.

6. **Monitor progress and adjust as needed:**

- As the project progresses, it is important to regularly review progress and adjust the plan as needed. This may involve making changes to the timeline or resource allocation.

7. **Communicate with stakeholders:**

- Communication is key throughout the entire project lifecycle. Stakeholders should be kept informed of progress, issues, and any changes to the plan.

8. **Evaluate performance and close the project:**

- At the end of the project, it is important to evaluate performance and document any lessons learned. This can help to identify areas that need improvement for future projects.

2. Establish the requirement for the project:

a. Human:

- **Project Manager** - - Danyil Tymchuk
- **Design** (UI/UX) - - Danyil Tymchuk
- **Graphic** (3d/2d) - - Danyil Tymchuk
- **Programing** (Unity 3d(with C#)) - - Danyil Tymchuk
- **Testing** (QA) - - Danyil Tymchuk

b. Hardware:

- **PC / Laptop:** For program development (I have)
- **System Requirements:**

- Windows:

Operating system version:

- Windows 7 (SP1+)
- Windows 10, 64-bit versions only
- Windows 11, 64-bit versions only

CPU:

- X64 architecture with SSE2 instruction set support

Graphics API:

- DX10
- DX11
- DX12-capable GPUs

Additional requirements:

- Hardware vendor officially supported drivers

- macOS:

Operating system version:

- High Sierra 10.13+ (Intel editor)
- Big Sur 11.0 (Apple silicon Editor)

CPU:

- X64 architecture with SSE2 instruction set support (Intel processors) Apple M1 or above (Apple silicon-based processors)

Graphics API:

- Metal-capable Intel and AMD GPUs

Additional requirements:

- Apple officially supported drivers (Intel processor)
- Rosetta 2 is required for Apple silicon devices running on either Apple silicon or Intel versions of the Unity Editor

- Linux:

- Operating system version:**

- Ubuntu 20.04
 - Ubuntu 18.04
 - CentOS 7

- CPU:**

- X64 architecture with SSE2 instruction set support

- Graphics API:**

- OpenGL 3.2+
 - Vulkan-capable
 - Nvidia
 - AMD GPUs.

- Additional requirements:**

- Gnome desktop environment running on top of X11 windowing system, Nvidi official proprietary graphics driver or AMD Mesa graphics driver. Other configuration and user environment as provided stock with the supported distribution (Kernel, Compositor, etc.)

- **Android phone:** For testing this program (I have):

- **System Requirements:**

- Android:

- Android 7.0 or newer
 - Bluetooth connectivity/Wi-Fi
 - Field of view in 3D view(camera)
 - Display capacity
 - Visual tracking
 - Digital projectors
 - Accelerometer
 - Gyroscope
 - Magnetometer

c. Software:

- **Unity 3d(with C#):**
 - IDE for game development, but we only use for development of AR program (free)
- **Figma:**
 - For the development of program design (free)
- **Blender:**
 - For development of 3d object (free)
- **Photoshop:**
 - For development of graphic objects (icons / 2d objects) (146\$/year)
- **Google Play Console:**
 - For publication in Google Play(25\$)

3. Use of correct documentation and content:

Agile development process:

- **Sprint 1:**

- **Sprint Goal:**

- The goal of this sprint is to lay the foundation for the project by setting up the project's infrastructure and creating the basic user interface

- **User Stories:**

- As a user, I want to be able to create an account and log in to the game
 - As a user, I want to be able to navigate through the game's main menu
 - As a user, I want to be able to view the game's tutorial

- **Tasks:**

- Set up the project's infrastructure (e.g. source control, development environment)
 - Create the game's main menu screen
 - Implement the account creation and login functionality
 - Create the tutorial screen

- **Deliverables:**

- A working login and main menu screen
 - A working tutorial screen
 - A functional account creation and login system

- **Sprint 2:**

- **Sprint Goal:**

- The goal of this sprint is to implement the core gameplay mechanics of the game

- **User Stories:**

- As a user, I want to be able to move my object around the game world
 - As a user, I want to be able to interact with objects in the game world
 - As a user, I want to be able to complete objectives and earn rewards

- **Tasks:**

- Implement the object movement mechanics
 - Implement object interaction mechanics
 - Implement the game's objective and reward system

- **Deliverables:**

- A working object movement system
 - A functional object interaction system
 - A functional objective and reward system

Trello

<https://trello.com/invite/b/4rlFjuBs/ATTI91fbc418e4ae84ee3f0592f3dcb4ba16FFF47F5C/project-management>

